

Hybrid Pre-Release Heavy Test Report

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Release recommendation: GO

Executive Scope and Caveats

Hybrid is a limited optimization scheme for bounded local workflows. It is not a universal optimization scheme for arbitrary general tasks.

The strategy can still be run on general tasks, but outside the validated task families in this report, neither quality parity nor token or time savings are guaranteed.

These results should therefore be read as a release-facing estimate for a narrow operating range, not as a guaranteed average for tasks of any type and any complexity.

Method

This formal pre-release run compares a natural-language baseline against the current adaptive hybrid strategy.

The suite contains 12 local heavy tasks: four tiers and three groups per tier.

Token consumption uses Codex-reported total tokens when available and falls back to estimated prompt plus visible output tokens only when the reported count is missing.

Quality uses task-specific validators normalized to a 0-100 score.

The summary also tracks contract family, adapter family, validator type, resumability coverage, and exactness preservation coverage so tier alone is not treated as the full semantic classifier.

Validation Dimensions

Resumability coverage: 3/3 tasks (100.00%).

Exactness preservation coverage: 9/12 tasks (75.00%).

Classification Limits

- The four-tier hybrid classification is a routing heuristic driven by runtime structure, not a universal taxonomy of task meaning or business value.
- The same user-facing request can fall into different tiers once it gains explicit files, reusable handoff state, loop checkpoints, or resume requirements.

- Mixed tasks are especially unstable at the boundaries. A task that begins as a short read-only request can become a multi-step repair or a loop-heavy workflow as soon as new acceptance checks appear during execution.
- The classification is weakest when the real success criteria live outside the prompt, outside the local workspace, or outside validator-checkable artifacts.

Recommended Task Families

- Local file reads or summaries with strict output shape and explicit literal requirements.
- One-pass review tasks with exactly one deliberate handoff and a concrete written artifact.
- Multi-step local code or config repairs with named files, named tests or commands, and explicit machine-readable handoff artifacts.
- Looped document or playbook rewrites that have checkpointable structure and objective content requirements.

Task Families That Should Not Use This Report as a Proxy

- Open-ended web research, product strategy, or exploratory planning where scope changes during the run.
- Creative generation, taste-driven writing, or tasks whose quality is mostly subjective rather than validator-based.
- Production operations, remote systems work, or workflows dominated by network latency, approvals, or external services.
- Human-in-the-loop tasks where missing tacit context matters more than the prompt contract or local artifact discipline.

Why T0 Can Show Negative Optimization

In this formal run, T0 as a tier shows -46.73% token optimization, 31.89% time optimization, and average quality improving from 95.00 to 100.00.

The negative token outliers are Three-line release notes summary (-80.30%), Routing threshold brief (-69.70%).

The main cause is fixed hybrid overhead. On tiny read-only tasks, routing wrappers, output-preservation guards, and a stricter prompt contract can cost more tokens than the task itself would otherwise consume.

Operationally, this does not indicate a quality failure in the current run. The T0 hybrid outputs matched or exceeded baseline quality, but efficiency became unreliable because the task bodies were too small to amortize the extra control cost.

Practical implication: use hybrid on T0-sized work when format fidelity, literal retention, or consistency matters. Do not expect reliable token savings on very small tasks.

Task Selection

Tier	Group	Title	Representative Task Description
t0	1	Three-line release notes summary	Read a short release-notes file and return exactly three non-empty lines. The answer must cite the source file path and include two named release facts, so this task represents a tiny read-only extraction with strict wording requirements.
t0	2	Routing threshold brief	Read a small routing-threshold reference and produce a fixed-format brief that cites the source path and the key threshold facts. This models a tiny configuration lookup where format fidelity matters more than synthesis depth.
t0	3	Operator note brief	Read a short operator note and extract a three-line brief with the required path and two explicit facts. This represents a very small read-only summarization task with literal evidence requirements.
t1	1	Empty-input bug review	Inspect a small Python function that fails on empty input, perform exactly one reviewer handoff, and write a structured review report with a decision and evidence. This simulates a lightweight one-pass review workflow.
t1	2	Zero-sample bug review	Inspect a latency helper that breaks when the sample count is zero, perform exactly one reviewer handoff, and deliver a structured review note. This tests a single coordination step without reusable runtime state.
t1	3	Missing-route bug review	Inspect a route selector that crashes on unknown keys, perform exactly one reviewer handoff, and write a structured review result. This is another single-handoff review task with concrete evidence requirements.
t2	1	Shared pipeline repair	Repair a local deduplication pipeline, run the named unit-test command, write a shared-state artifact for the next phase, and produce review notes. This represents a multi-step engineering fix that has both code and handoff artifacts.
t2	2	Reviewer queue repair	Fix a queue-normalization bug, run the named unit-test command, write the shared-state handoff artifact, and leave a structured review note. This models a multi-step local fix with reusable machine-readable state.
t2	3	Stage plan repair	Restore a missing first stage in a local deployment-plan workflow, rerun unit tests, write the shared-state artifact for the next phase, and produce review notes. This is a multi-step repair where task-contract fidelity matters because several artifacts must stay aligned.
t3	1	Review loop playbook rewrite	Rewrite a review playbook so each review cycle stays evidence-first, roles remain explicit, scope boundaries are preserved, and checkpoint notes keep the loop state recoverable. This is a loop-heavy document task rather than a code-fix task.
t3	2	Release signoff handbook rewrite	Rewrite a release-signoff handbook so blocker-first checkpoints remain intact before final signoff and the document can be resumed cleanly from saved progress notes. This represents a loop-heavy procedural document task.
t3	3	Resume handoff guide rewrite	Rewrite a resume-and-handoff guide so it preserves latest-checkpoint recovery, continues with unresolved items first, and keeps the recovery loop consistent across restarts. This is a loop-heavy continuity document task.

Per-Task Metrics

Tier	Group	Baseline Tokens	Hybrid Tokens	Token Opt %	Baseline Time (s)	Hybrid Time (s)	Time Opt %	Baseline Quality	Hybrid Quality
t0	1	3320	5986	-80.30%	19.15	23.73	-23.93%	85	100
t0	2	3541	6009	-69.70%	35.61	11.51	67.67%	100	100
t0	3	3365	3010	10.55%	18.49	14.65	20.79%	100	100
t1	1	145690	13238	90.91%	367.44	42.46	88.45%	100	100
t1	2	5866	16279	-177.51%	336.45	95.65	71.57%	100	100
t1	3	20236	11584	42.76%	139.29	34.55	75.20%	100	100
t2	1	33823	7342	78.29%	208.84	61.57	70.52%	85	100
t2	2	24039	11570	51.87%	139.66	43.35	68.96%	85	100
t2	3	48130	8251	82.86%	147.95	39.24	73.48%	85	100
t3	1	7308	8145	-11.45%	79.26	47.44	40.15%	100	100
t3	2	17817	8487	52.37%	118.22	45.19	61.78%	100	100
t3	3	8283	8264	0.23%	124.65	41.03	67.08%	100	100

Tier Aggregates

Tier	Task Count	Baseline Tokens	Hybrid Tokens	Token Opt %	Baseline Time (s)	Hybrid Time (s)	Time Opt %	Baseline Avg Quality	Hybrid Avg Quality
t0	3	10226	15005	-46.73%	73.24	49.89	31.89%	95.00	100.00
t1	3	171792	41101	76.08%	843.19	172.66	79.52%	100.00	100.00
t2	3	105992	27163	74.37%	496.45	144.16	70.96%	85.00	100.00
t3	3	33408	24896	25.48%	322.13	133.66	58.51%	100.00	100.00

Detailed Findings

t0 G1 - Three-line release notes summary

Task description: Read a short release-notes file and return exactly three non-empty lines. The answer must cite the source file path and include two named release facts, so this task represents a tiny read-only extraction with strict wording requirements.

Contract family: read_extract

Adapter family: docs

Validator type: exact_output

Observed adapter id: generic-fs+docs

Contract hash: 8f1dbbe05f2a1134812cafdd12a2975e206d12dfc30031194b4acb3b28da0a5f

NL baseline tokens: 3320

Hybrid tokens: 5986

Token optimization pct: -80.30%

NL baseline runtime seconds: 19.15

Hybrid runtime seconds: 23.73

Time optimization pct: -23.93%

NL baseline quality: 85 (good)

Hybrid quality: 100 (excellent)

Quality delta: 15

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: exactly 3 non-empty lines; tier line correct; evidence names docs/release_notes.txt; required facts present: 1/2

Hybrid notes: exactly 3 non-empty lines; tier line correct; evidence names docs/release_notes.txt; required facts present: 2/2

t0 G2 - Routing threshold brief

Task description: Read a small routing-threshold reference and produce a fixed-format brief that cites the source path and the key threshold facts. This models a tiny configuration lookup where format fidelity matters more than synthesis depth.

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Contract family: generic_config_extract

Adapter family: generic-fs

Validator type: exact_output

Observed adapter id: generic-fs

Contract hash: e951e5e6ce719d4f81a3700c0ae8b4176ec47a5b1887d4bc7e87d5d26188ea08

NL baseline tokens: 3541

Hybrid tokens: 6009

Token optimization pct: -69.70%

NL baseline runtime seconds: 35.61

Hybrid runtime seconds: 11.51

Time optimization pct: 67.67%

NL baseline quality: 100 (excellent)

Hybrid quality: 100 (excellent)

Quality delta: 0

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: exactly 3 non-empty lines; tier line correct; evidence names data/router_thresholds.json; required facts present: 2/2

Hybrid notes: exactly 3 non-empty lines; tier line correct; evidence names data/router_thresholds.json; required facts present: 2/2

t0 G3 - Operator note brief

Task description: Read a short operator note and extract a three-line brief with the required path and two explicit facts. This represents a very small read-only summarization task with literal evidence requirements.

Contract family: read_extract

Adapter family: docs

Validator type: exact_output

Observed adapter id: generic-fs+docs

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Contract hash: adbcff351b60512f3a9c0f5ae39d36640c942d5287f10bf69ed6cfff61adbf39

NL baseline tokens: 3365

Hybrid tokens: 3010

Token optimization pct: 10.55%

NL baseline runtime seconds: 18.49

Hybrid runtime seconds: 14.65

Time optimization pct: 20.79%

NL baseline quality: 100 (excellent)

Hybrid quality: 100 (excellent)

Quality delta: 0

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: exactly 3 non-empty lines; tier line correct; evidence names docs/operator_notes.txt; required facts present: 2/2

Hybrid notes: exactly 3 non-empty lines; tier line correct; evidence names docs/operator_notes.txt; required facts present: 2/2

t1 G1 - Empty-input bug review

Task description: Inspect a small Python function that fails on empty input, perform exactly one reviewer handoff, and write a structured review report with a decision and evidence. This simulates a lightweight one-pass review workflow.

Contract family: single_handoff_review

Adapter family: python

Validator type: structured_review

Observed adapter id: generic-fs+python+docs

Contract hash: bf9d1aaf66223a865b0a366f9aa13da317dae8a02d78e4fc925e18f89f62babf

NL baseline tokens: 145690

Hybrid tokens: 13238

Token optimization pct: 90.91%

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NL baseline runtime seconds: 367.44

Hybrid runtime seconds: 42.46

Time optimization pct: 88.45%

NL baseline quality: 100 (excellent)

Hybrid quality: 100 (excellent)

Quality delta: 0

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: review report written; required headings present; report cites src/review_target.py; bug explanation checks: 2/2

Hybrid notes: review report written; required headings present; report cites src/review_target.py; bug explanation checks: 2/2

t1 G2 - Zero-sample bug review

Task description: Inspect a latency helper that breaks when the sample count is zero, perform exactly one reviewer handoff, and deliver a structured review note. This tests a single coordination step without reusable runtime state.

Contract family: single_handoff_review

Adapter family: python

Validator type: structured_review

Observed adapter id: generic-fs+python+docs

Contract hash: 5c952a94a25721d9356c4dbdca13866327eb3be55997f4bd6fc1c6893f883a69

NL baseline tokens: 5866

Hybrid tokens: 16279

Token optimization pct: -177.51%

NL baseline runtime seconds: 336.45

Hybrid runtime seconds: 95.65

Time optimization pct: 71.57%

NL baseline quality: 100 (excellent)

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Hybrid quality: 100 (excellent)

Quality delta: 0

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: review report written; required headings present; report cites src/latency_target.py; bug explanation checks: 2/2

Hybrid notes: review report written; required headings present; report cites src/latency_target.py; bug explanation checks: 2/2

t1 G3 - Missing-route bug review

Task description: Inspect a route selector that crashes on unknown keys, perform exactly one reviewer handoff, and write a structured review result. This is another single-handoff review task with concrete evidence requirements.

Contract family: single_handoff_review

Adapter family: python

Validator type: structured_review

Observed adapter id: generic-fs+python+docs

Contract hash: e972cc9c18b11ceeb862de3e26c4b47e8bb0c053b528f4186cf44abc05e962c0

NL baseline tokens: 20236

Hybrid tokens: 11584

Token optimization pct: 42.76%

NL baseline runtime seconds: 139.29

Hybrid runtime seconds: 34.55

Time optimization pct: 75.20%

NL baseline quality: 100 (excellent)

Hybrid quality: 100 (excellent)

Quality delta: 0

Route match: True

Adapter match: True

Hybrid Pre-Release Heavy Test Report

Resumability covered: True

Exactness preserved: True

Baseline notes: review report written; required headings present; report cites src/router_target.py; bug explanation checks: 2/2

Hybrid notes: review report written; required headings present; report cites src/router_target.py; bug explanation checks: 2/2

t2 G1 - Shared pipeline repair

Task description: Repair a local deduplication pipeline, run the named unit-test command, write a shared-state artifact for the next phase, and produce review notes. This represents a multi-step engineering fix that has both code and handoff artifacts.

Contract family: shared_state_repair

Adapter family: python

Validator type: unit_tests_and_artifacts

Observed adapter id: generic-fs+python+docs

Contract hash: 77ef1e2c56e0f8f6ccf3433ee7ed4cd364670f5ddcc11806e2f9f37b64695bb4

NL baseline tokens: 33823

Hybrid tokens: 7342

Token optimization pct: 78.29%

NL baseline runtime seconds: 208.84

Hybrid runtime seconds: 61.57

Time optimization pct: 70.52%

NL baseline quality: 85 (good)

Hybrid quality: 100 (excellent)

Quality delta: 15

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: False

Baseline notes: unit tests passed; shared state ACL-X artifact missing or invalid; review notes present

Hybrid notes: unit tests passed; shared state ACL-X artifact present; review notes present

t2 G2 - Reviewer queue repair

Task description: Fix a queue-normalization bug, run the named unit-test command, write the shared-state handoff artifact, and leave a structured review note. This models a multi-step local fix with reusable machine-readable state.

Contract family: shared_state_repair

Adapter family: python

Validator type: unit_tests_and_artifacts

Observed adapter id: generic-fs+python+docs

Contract hash: 04f99d71b955bc009f52d05fee0835f32965a60e200cd9934a9aab62e0f24e83

NL baseline tokens: 24039

Hybrid tokens: 11570

Token optimization pct: 51.87%

NL baseline runtime seconds: 139.66

Hybrid runtime seconds: 43.35

Time optimization pct: 68.96%

NL baseline quality: 85 (good)

Hybrid quality: 100 (excellent)

Quality delta: 15

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: False

Baseline notes: unit tests passed; shared state ACL-X artifact missing or invalid; review notes present

Hybrid notes: unit tests passed; shared state ACL-X artifact present; review notes present

t2 G3 - Stage plan repair

Task description: Restore a missing first stage in a local deployment-plan workflow, rerun unit tests, write the shared-state artifact for the next phase, and produce review notes. This is a multi-step repair where task-contract fidelity matters because several artifacts must stay aligned.

Contract family: shared_state_repair

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Adapter family: python

Validator type: unit_tests_and_artifacts

Observed adapter id: generic-fs+python+docs

Contract hash: 9ffe2f06a7a9d91225066d409ff272f08fabd01dba0b2e8a885bf6443d5a87d1

NL baseline tokens: 48130

Hybrid tokens: 8251

Token optimization pct: 82.86%

NL baseline runtime seconds: 147.95

Hybrid runtime seconds: 39.24

Time optimization pct: 73.48%

NL baseline quality: 85 (good)

Hybrid quality: 100 (excellent)

Quality delta: 15

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: False

Baseline notes: unit tests passed; shared state ACL-X artifact missing or invalid; review notes present

Hybrid notes: unit tests passed; shared state ACL-X artifact present; review notes present

t3 G1 - Review loop playbook rewrite

Task description: Rewrite a review playbook so each review cycle stays evidence-first, roles remain explicit, scope boundaries are preserved, and checkpoint notes keep the loop state recoverable. This is a loop-heavy document task rather than a code-fix task.

Contract family: loop_document_rewrite

Adapter family: docs

Validator type: document_exactness_and_checkpoint

Observed adapter id: generic-fs+docs

Contract hash: c856537d4b528f22cfa9d4646153d3ae6eab9737850a7dcec7d01baf7f408a70

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NL baseline tokens: 7308

Hybrid tokens: 8145

Token optimization pct: -11.45%

NL baseline runtime seconds: 79.26

Hybrid runtime seconds: 47.44

Time optimization pct: 40.15%

NL baseline quality: 100 (excellent)

Hybrid quality: 100 (excellent)

Quality delta: 0

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: document content checks: 9/9; document heading checks: 4/4; checkpoint note checks: 3/3

Hybrid notes: document content checks: 9/9; document heading checks: 4/4; checkpoint note checks: 3/3

t3 G2 - Release signoff handbook rewrite

Task description: Rewrite a release-signoff handbook so blocker-first checkpoints remain intact before final signoff and the document can be resumed cleanly from saved progress notes. This represents a loop-heavy procedural document task.

Contract family: loop_document_rewrite

Adapter family: docs

Validator type: document_exactness_and_checkpoint

Observed adapter id: generic-fs+docs

Contract hash: 4f570db97da561e87d6aad46aaae94ac341fb58177263285e63ca32d461ea433

NL baseline tokens: 17817

Hybrid tokens: 8487

Token optimization pct: 52.37%

NL baseline runtime seconds: 118.22

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Hybrid runtime seconds: 45.19

Time optimization pct: 61.78%

NL baseline quality: 100 (excellent)

Hybrid quality: 100 (excellent)

Quality delta: 0

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: document content checks: 8/8; document heading checks: 4/4; checkpoint note checks: 3/3

Hybrid notes: document content checks: 8/8; document heading checks: 4/4; checkpoint note checks: 3/3

t3 G3 - Resume handoff guide rewrite

Task description: Rewrite a resume-and-handoff guide so it preserves latest-checkpoint recovery, continues with unresolved items first, and keeps the recovery loop consistent across restarts. This is a loop-heavy continuity document task.

Contract family: loop_document_rewrite

Adapter family: docs

Validator type: document_exactness_and_checkpoint

Observed adapter id: generic-fs+docs

Contract hash: 012887b1708085c7226d71aed20fd6f240fbde36ef1d693bd1d3a7567a820094

NL baseline tokens: 8283

Hybrid tokens: 8264

Token optimization pct: 0.23%

NL baseline runtime seconds: 124.65

Hybrid runtime seconds: 41.03

Time optimization pct: 67.08%

NL baseline quality: 100 (excellent)

Hybrid quality: 100 (excellent)

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Quality delta: 0

Route match: True

Adapter match: True

Resumability covered: True

Exactness preserved: True

Baseline notes: document content checks: 9/9; document heading checks: 4/4; checkpoint note checks: 3/3

Hybrid notes: document content checks: 9/9; document heading checks: 4/4; checkpoint note checks: 3/3

Release Decision

Baseline total tokens: 321418

Hybrid total tokens: 108165

Token optimization pct: 66.35%

Baseline total seconds: 1735.01

Hybrid total seconds: 500.36

Time optimization pct: 71.16%

Baseline average quality: 95.00

Hybrid average quality: 100.00

Quality delta: 5.00

Recommendation: GO for the bounded local engineering range covered by this report. Gate checks passed, the strategy lock stayed clean, hybrid routed to the expected tier, and the average hybrid quality did not fall below the baseline.

This recommendation does not convert hybrid into a general-purpose optimization guarantee for arbitrary tasks.